

Introduction to Cloud Computing

This resource is based on definitions from **NIST** (the U.S. National Institute of Standards and Technology), the organization whose definitions are the industry-standard starting point for describing cloud computing. Where a word might be new or tricky, a plain-English version is given in *[brackets]* right after it.

1. What is Cloud Computing?

Cloud computing is the **practice of using the internet to access a set of remote servers** *[computers owned and run by someone else, in a data centre far away]* to **store, manage, and process data**.

Key idea: you use it **on demand** *[whenever you need it, without asking anyone first]* and **pay as you go** *[you only pay for what you actually use, like an electricity or water bill]*.

2. The 5 Major Cloud Actors

(Source: NIST / ISO — the two standards bodies that define these roles)

These are the five "characters" involved in any cloud computing relationship.

1. Cloud Provider

- Owns the **data centres** *[physical buildings full of servers]*.
- **Supplies the services** (e.g., storage, computing power) to others.
- Is an **organization or entity** — usually a company like Amazon, Microsoft, or Google.

2. Cloud Consumer

- The **customer**.
- The **principal user** *[main person or organization actually using the service]*.
- Has a **business relationship with the provider** (e.g., a contract, a subscription, a pay-as-you-go account).

3. Cloud Broker

- An entity that **manages the use, performance, and delivery** of cloud services on behalf of a consumer.
- **Negotiates the relationship** between the provider and the consumer.
- Example: **consulting firms** that help a business choose and manage the right cloud services for them.

4. Cloud Auditor

- Conducts an **independent assessment** *[an outside, unbiased check-up]* of cloud services.
- Mainly concerned with **privacy and security**.
- Example: a **security firm**, or checking compliance with laws like **GDPR** *[General Data Protection Regulation — the EU's data privacy law]*.

5. Cloud Carrier

- The **intermediary** [*the "middleman" or connector*] that provides **connectivity and transport** of cloud services between the provider and the consumer.
- In practice, this is usually an **ISP** [*Internet Service Provider*] — for example, Telenor.

How the Actors Relate to Each Other

- The **Provider** and **Consumer** have the core business relationship.
- The **Broker** sits between them, managing that relationship.
- The **Auditor** independently checks the Provider (privacy/security compliance).
- The **Carrier** is the network connection making the whole relationship possible.

3. The 5 Essential Characteristics of Cloud Computing

(Source: NIST — csrc.nist.gov, nvlpubs.nist.gov)

These are the five features that make a service "cloud computing." If a service is missing one of these, it probably isn't truly cloud.

1. On-Demand Self-Service

- The consumer [*customer*] can **provision** [*set up / spin up*] computing power (like storage or servers) automatically, **without needing to talk to a human** at the provider.
- This is often called "**click and go**."
- Related term: **IaC — Infrastructure as Code** [*setting up computer systems using written scripts instead of manually clicking through menus, so it can be automated*].

2. Broad Network Access

- Cloud capabilities are available **over the internet**, from almost anywhere.
- You can access services using **standard tools** [*a web browser, an app, or a command line*].
- You can connect from **heterogeneous platforms** [*different types of devices — a phone, a laptop, a workstation*].

3. Resource Pooling

- The provider's hardware is **shared between many different customers** at the same time — this is called **multi-tenant**.
- **Physical and virtual resources** [*real hardware, and software-simulated versions of hardware*] are assigned dynamically based on this multi-tenant model.
- Even though resources are shared, your data stays **isolated** [*kept separate and private from other tenants*].
- The customer usually **doesn't know or care which physical machine** their data is actually running on — this is called **location independence**.

4. Rapid Elasticity

- The system's capabilities can **grow and shrink automatically to match demand** — like a rubber band stretching and relaxing.
- Example: a website might need very little computing power at 7 AM, a lot at lunchtime, and taper off by 7 PM. The cloud stretches and shrinks to follow this pattern automatically.

- To the consumer, this capacity **appears unlimited** — you never "run out."

5. Measured Service

- Cloud systems **automatically track and control how much you use** — like a **utility meter** *[the box on your house that tracks how much electricity or water you use, so you get billed accurately]*.
- Different services are measured with **different metrics** *[units of measurement]*, for example:
 - **Storage** → measured in **GB** *[gigabytes]*
 - **Virtual Machines (VMs)** → measured in **CPU hours** *[how long and how much processing power was used]*
- This usage data is visible to **both the provider and the consumer**, so both sides can see (and be billed/charged) transparently.

4. Sources

This material is based on definitions published by NIST and related standards references:

1. csrc.nist.gov
2. nvlpubs.nist.gov
3. csrc.nist.gov
4. peasoup.cloud
5. synopsys.com
6. cst.gov.sa
7. noraonline.nl